

Davide Scassola

PHD IN AI · RESEARCH SCIENTIST

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Applied scientist specializing in probability and AI with experience in both academic and industrial settings. Experienced across the full lifecycle, from solving statistical, probabilistic, and algorithmic problems to implementing reliable software.

Work Experience

Aindo SpA Research Scientist • Developed and tested a novel privacy metric for synthetic data and benchmarked existing privacy metrics. • Conducted research on tabular and relational data generation (PhD collaboration).	<i>Trieste, Italy</i> <i>Jan. 2023 - May 2026</i>
School of Informatics, University of Edinburgh Visiting Researcher • Conducted research on tabular data generation with probabilistic circuits. Supervisor: Antonio Vergari.	<i>Edinburgh, UK</i> <i>May. 2025 - Oct. 2025</i>
Trality GmbH R&D Scientist, Backend Developer (quantitative finance) • Back-end development: Implemented trading strategy optimizer, statistical indicators, rebalancing of margin-trading portfolios. • Research: Conducted research on cryptocurrency time series prediction models, reinforcement learning for cryptocurrency trading, performance metrics for trading strategies.	<i>Vienna, Austria</i> <i>May. 2021 - Dec. 2022</i>
Deloitte Consulting IT Analyst • Performed statistical analysis of banking datasets and defined automated data quality controls. • Analyzed data governance solutions in the banking sector.	<i>Milan, Italy</i> <i>Feb. 2021 - Apr. 2021</i>
Aindo SpA Research Intern • Developed a deep generative model for multivariate financial time series that simulates real dynamics and statistical properties.	<i>Trieste, Italy</i> <i>Sept. 2019 - Dec. 2019</i>

Education

University of Trieste Doctorate (PhD) in Applied Data Science and Artificial Intelligence · Graduated Cum Laude Developed new techniques for tabular data generation based on diffusion models, graph neural networks, neuro-symbolic constraints and probabilistic circuits. PhD in collaboration with Aindo SpA. • Thesis: <i>Towards Flexible and Expressive Generative Models for Tabular and Relational Data</i> . • Advisor: Luca Bortolussi (University of Trieste), co-advisor: Sebastiano Sacconi (Aindo SpA) • Teaching assistant for the Master's course in Probabilistic Machine Learning (2024). • Reviewer for the AAAI 2026 conference.	<i>Trieste, Italy</i> <i>Jan. 2023 - Dec. 2025</i>
University of Trieste, International School for Advanced Studies (SISSA) Master of Science (MSc) in Data Science and Scientific Computing · Graduation Score: 110 Cum Laude • Thesis: <i>An Information-Theoretic Investigation Into Epigenetic Regulation of Gene Expression</i> . • Advisors: Guido Sanguinetti (SISSA), Matteo Marsili (ICTP) • Artificial Intelligence and Machine Learning track, all courses taught and assessed in English.	<i>Trieste, Italy</i> <i>Oct. 2018 - Dec. 2020</i>
University of Brescia Bachelor of Engineering (BEng) in Computer Science and Engineering · Graduation Score: 110 Cum Laude	<i>Brescia, Italy</i> <i>Sept. 2015 - Sept. 2018</i>

Skills & Interests

Advanced	Python
Some Experience	C, C++, Java, R, SQL
Other tools	Data science and machine learning software ecosystem (NumPy, SciPy, Pandas, scikit-learn), PyTorch, Pyro, Git, LaTeX, shell scripting, Google Workspace, Microsoft Office Suite
Areas of Interest	Probabilistic models, Machine Learning & AI, Software Engineering
Languages	English (fluent, certified C1), Italian (native)

Honors & Awards

2024 **Best Poster**, Eastern European Machine Learning Summer School (EEML)

Novi Sad, Serbia

2018 **Data Science Excellence Fellowship (12k€)**, International School of Advanced Studies (SISSA)

Trieste, Italy

Publications

ReMIA: a Powerful and Efficient Alternative to Membership Inference Attacks against Synthetic Data Generators

arXiv

Scassola D., Coser A., Saccani S.

2026

A Sobering Look at Tabular Data Generation via Probabilistic Circuits

UAI

Scassola D., Ponsford D., Javaloy A., Saccani S., Bortolussi L., Gouk H., Vergari A.

2026

Graph-Conditional Flow Matching for Relational Data Generation

AAAI

Scassola D., Saccani S., Bortolussi L.

2026

Zero-Shot Conditioning of Score-Based Diffusion Models by Neuro-Symbolic Constraints

AAAI

Scassola D., Carbone G., Saccani S., Bortolussi L.

2025

Multi-Objective Reward Generalization: Improving Performance of Deep Reinforcement Learning for Selected Applications in Stock and Cryptocurrency Trading

Neural Computing & Applications

Cornalba F., Disselkamp C., Scassola D., Helf C.

2024

Model Abstraction and Conditional Sampling with Score-Based Diffusion Models

QEST

Bortolussi L., Cairoli F., Giacomarra F., Scassola D.

2023